

T20 Cricket Player Analytics Dashboard

INTRODUCTION

Cricket is one of the most popular sports in the world, and T20 cricket is known for its fast-paced and exciting format. In this project, I developed a T20 Cricket Player Analytics Dashboard using Power BI to analyze the batting performance of players from different countries.

The purpose of this dashboard is to provide meaningful insights about player performance by using different visualizations and key performance indicators (KPIs). The dashboard helps users understand player statistics such as runs scored, batting average, strike rate, centuries, fifties, and sixes.

Objective

The main objectives of this project are:

- To analyze the batting performance of T20 cricket players.
 - To identify the top run scorers.
 - To identify players who have hit the most sixes.
 - To compare player performance using batting average and strike rate.
 - To perform country-wise analysis of players.
 - To create an interactive dashboard using Power BI.
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Tools and Technologies Used

The following tools were used to complete this project:

- Microsoft Power BI Desktop
- Power Query Editor
- DAX (Data Analysis Expressions)
- Microsoft Excel

Dataset Description

The dataset used in this project contains batting statistics of T20 international cricket players. It includes information such as player names, country, matches played, innings, runs scored, strike rate, centuries, fifties, fours, sixes, and batting averages.

Some important columns used in the project are:

- Player
- Country
- Matches (Mat)
- Innings (Inns)
- Not Outs (NO)
- Runs
- Highest Score (HS)
- Strike Rate (SR)
- Centuries (100)
- Fifties (50)
- Fours (4s)
- Sixes (6s)
- Start Year
- End Year

The dataset contains more than 2000 player records.

Data Cleaning and Transformation

Before creating the dashboard, the dataset was cleaned and transformed using Power Query Editor.

The following steps were performed:

- Checked the dataset for missing values.
- Converted columns into appropriate data types.
- Split the Span column into Start Year and End Year.
- Standardized country names for better analysis.
- Created a Country_Clean column to remove inconsistencies in country values.
- Verified numerical columns before creating calculations and visuals.

These steps helped improve data quality and accuracy.

Dashboard Development

The dashboard was designed to provide a clear overview of player performance.

The following KPIs were created:

Total Players

Shows the total number of players available in the dataset.

Total Runs

Shows the total runs scored by all players.

Total Fifties

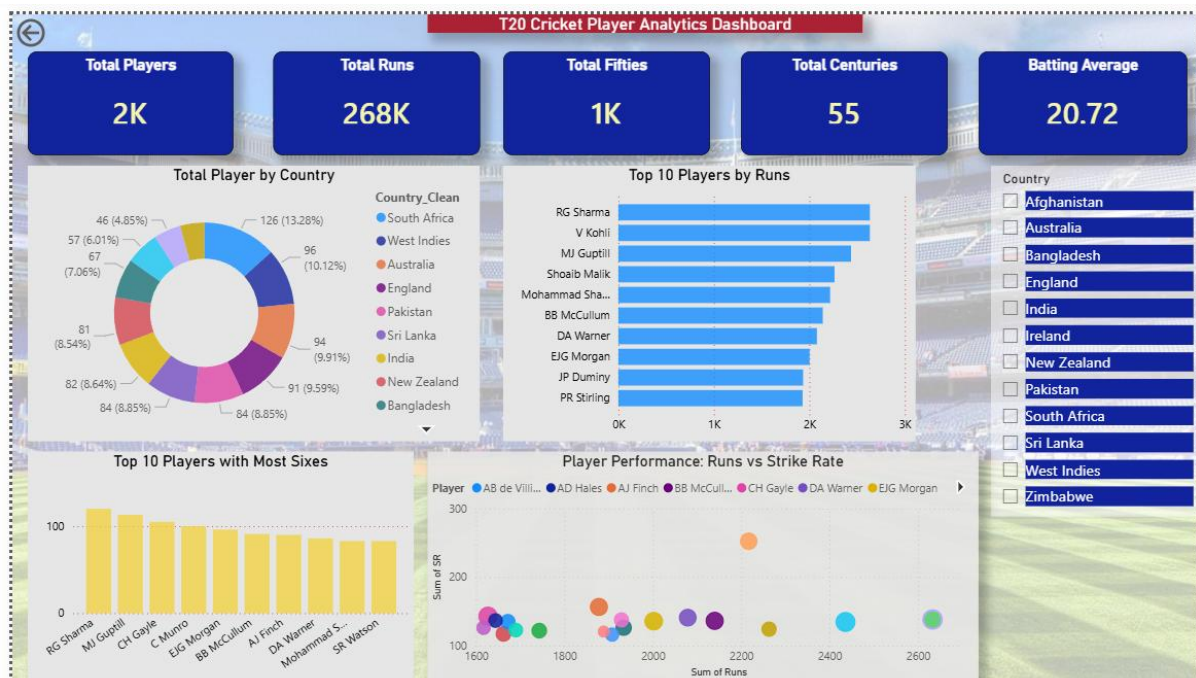
Shows the total number of half-centuries scored by players.

Total Centuries

Shows the total number of centuries scored by players.

Batting Average

Shows the average batting performance of players based on runs scored and dismissals.



Visualizations Used

Country-wise Player Distribution

A donut chart was used to show the distribution of players from different cricket-playing countries.

This visual helps compare player participation among countries.

Top 10 Players by Runs

A bar chart was created to display the top 10 run scorers in T20 cricket.

This visual helps identify the most successful batsmen in terms of runs scored.

Top 10 Players with Most Sixes

A column chart was used to show the players who have hit the highest number of sixes.

This helps identify aggressive and power-hitting batsmen.

Runs vs Strike Rate Analysis

A scatter plot was created to analyze the relationship between runs scored and strike rate.

This visual helps identify players who score consistently while maintaining an aggressive batting approach.

Country Slicer

A country slicer was added to make the dashboard interactive.

Users can select a country and instantly view player statistics related to that country.

Key Insights from the Dashboard

After analyzing the dashboard, the following insights were observed:

- Rohit Sharma and Virat Kohli are among the leading run scorers in T20 cricket.
- Several players have maintained both high run totals and strong strike rates.
- The Top 10 Sixes chart highlights players known for aggressive batting.
- Countries such as South Africa, West Indies, Australia, England, and Pakistan contribute a significant number of players to the dataset.
- The scatter plot helps identify players who balance consistency and attacking gameplay.

Challenges Faced

While working on this project, a few challenges were encountered:

- Some country names were inconsistent and required cleaning.
- Certain columns contained missing values.
- Creating the correct batting average calculation required understanding cricket statistics.
- Designing a dashboard that was both informative and visually appealing required multiple revisions.

These challenges were solved through data cleaning, DAX calculations, and dashboard formatting.

Learning Outcomes

This project helped me improve my skills in:

- Data Cleaning using Power Query
- Creating KPIs using DAX
- Data Visualization in Power BI
- Dashboard Design
- Data Analysis and Interpretation
- Interactive Reporting

Conclusion

The T20 Cricket Player Analytics Dashboard successfully provides a clear analysis of player performance using Power BI. The dashboard allows users to explore batting statistics, compare players, and analyze country-wise contributions through interactive visuals.

Through this project, I gained practical experience in Power BI, Power Query, and DAX. The dashboard helped transform raw cricket data into meaningful insights and improved my understanding of data visualization and business intelligence concepts.

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